

Joshua Aney

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EDUCATION

Montana State University

Bachelor of Science, Computer Science, GPA: 3.96

Bozeman, MT

May 2026

SKILLS & COURSEWORK

Languages: Python, MATLAB, Java, C, SQL, C#

Frameworks: PyTorch, TensorFlow, WebDataset, .NET Core, SLURM, NumPy, Matplotlib

Tools: Linux, JetBrains Products, GitHub, VSCode, Adobe Illustrator, Overleaf, Claude Code

Coursework: Machine Learning, Linear Algebra, Artificial Intelligence, Greedy Algorithms, Flow Networks, Optimization, Linear and Integer Linear Programming, Technical Writing

EXPERIENCE

Research Technologist

June 2026 - Current, (Intern Summers 2024, 2025)

CaMP Lab (Cybernetics and Motor Physiology), Mayo Clinic

Rochester, MN

- **Tools:** MATLAB, Signal Processing, Adobe Illustrator, Linear Algebra, Technical Writing
- Conducted algorithmic analysis on human SEEG data to study how the brain responds to auditory stimuli.
- Implemented signal processing pipelines including noise reduction, baseline correction, and artifact rejection.
- Applied and modified Canonical Response Parameterization to allow for better temporal feature extraction when applied to auditory evoked responses.
- Created a new method for dynamic brain network identification using evoked responses.
- Working on guide for general sEEG preprocessing and referencing.
- Produced a poster for the 2026 BRAIN Initiative Conference - Aney, J., Jensen, M., Kerezoudis, P., Canales-Johnson, A., Hermes, D., Baker, M., Miller, K. *Parameterization and discovery of auditory response networks during an auditory oddball paradigm*

Undergraduate Research Assistant

January 2026 - Current

HACR Lab (Harnessing Automation in Cybersecurity Reasoning), Montana State University

Bozeman, MT

- **Tools:** Pytorch, Ghidra, High Performance Computing, SLURM
- Build pipelines to produce control flow graphs from compiled binary packages for code embedding generation.
- Design Graph Neural Network architectures using PyTorch based on recent research publications.
- Run machine learning workloads on Tempest, Montana State University's high performance computing system.

TECHNICAL PROJECTS

Machine Learning and AI Models from Scratch | *Python, NumPy, Linear Algebra, Latex*

2024-2026

- Implemented the following machine learning algorithms from scratch: Naïve Bayes, K-Nearest Neighbors, K-Means, and DBSCAN, Multi-Layer-Perceptron, Q-Learning, SARSA, and Value Iteration
- Implemented a First-Order-Logic engine to perform inference on knowledge bases using resolution.
- Implemented the following optimization algorithms from scratch: Backpropagation, Genetic Algorithm, Differential Evolution, Particle Swarm Optimization, and Simulated Annealing

Bozeman Bike Trails Application | *C#, Razor Pages, SQL Server, Mapbox*

2024

- Developed a web application to track user-riden bike trail GPS data

Hand Tracking / ASL Alphabet Detection | *Python, TensorFlow, MediaPipe*

2024

- Created a pipeline to collect data, train a dense neural network, and provide real time predictions for video inputs.

EXTRACURRICULAR ACTIVITIES & LEADERSHIP

President

Montana State University Tennis Club

August 2023 – Present

Bozeman, MT

Section Leader

Montana State University Drumline

April 2025 – January 2026

Bozeman, MT